

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
9	(a)			At least 1 wave in shallow water joining correctly to a deep water wave (1) A minimum of 3 wavefronts shown perpendicular to wave direction by eye (1) Smaller wavelength in shallow water must be consistent and a minimum of 3 wavefronts shown (1)		3		3		3
	(b)	(i)	I	1.5		1		1		1
			II	10 [cm] ecf		1		1	1	1
		(ii)		Wavelength is less in region B (1) [Frequency is constant] so wave speed is also less in region B, <u>so disagree</u> (1) Alternative: {The waves are closer / there are more waves} in region B (1) because the waves have slowed down in the shallow water, <u>so disagree</u> (1)			2	2		2
	(c)	(i)		6 [cm]		1		1		1
		(ii)		Selection and substitution: $\frac{75}{50}$ (1) = 1.5 [cm] (1)		2		2	2	2

		(iii)	A quarter of 60 is 15 (1) so increases to 75 [cm/s] (1) A quarter of 75 is 18.75 so should increase to 93.75 cm/s [but it increases to 82 cm/s] (1) [so the rule is not generally true and Janet is not correct] Alternative: Increase = 75 - 60 = 15 [cm/s] (1) So $\frac{15}{60} = \frac{1}{4}$ (1) Increase = 82 - 75 = 7 [cm/s], so $\frac{7}{75} \left[\neq \frac{1}{4} \right]$ (1) [so the rule is not generally true and Janet is not correct] Alternative: 60×1.25 (1) $= 75$ [cm/s] (1) $75 \times 1.25 = 93.75$ [cm/s] (1) [so the rule is not generally true and Janet is not correct] N.B. 2 marks can be awarded for the first calculation comparison of wave speeds at 8 and 6 [cm] or at 6 and 4 [cm] and then 1 mark for a second calculation comparison.			3	3	3	3
			Question 9 total	0	8	5	13	6	13